Joker of Academics

Measuring Imbalanced Emotional Investment: Construction and Application of a "Joker Index"

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Abstract: In recent years, "The joker turned out to be me" has become a popular metaphor in cyberspace for expressing imbalanced emotional investment, referring specifically to individuals who unilaterally over-invest in intimate or quasi-intimate relationships without receiving commensurate returns. This study aims to academicize this subcultural concept by constructing a quantifiable "Joker Index" determination model. Based on real interaction data extracted from WeChat chat log analysis tools, taking seven cases as examples, we identify three core dimensions: investment intensity, interaction tone gap, and emotional rumination, and process the raw indicators using multiple methods. Through multiple linear regression fitting of users' subjective ratings, the coefficients of the Joker Index formula are empirically determined. The model performs well on the seven cases, effectively distinguishing the identity and degree of the "joker" in relationships. This study provides a new quantitative tool for the sociology of emotion and internet linguistics research, and also helps individuals rationally examine their own emotional patterns.

Keywords: Joker Index; emotional imbalance; chat log analysis;multi-case study

1. Introduction

1.1 Research Origin

"In the end, the joker turned out to be me" — this internet meme has frequently appeared on social media in recent years, becoming a specific symbol for young people to express their emotional predicaments. In the context of popular culture, "joker" no longer merely refers to the comedic character in a circus, but has evolved to describe individuals who unilaterally over-invest in romantic or quasi-romantic relationships without receiving commensurate returns [1]. Behind this metaphor lies a common anxiety in contemporary intimate relationships: the imbalance between giving and receiving, the one-way flow of emotional investment, and the eventual absurdity of self-awakening [2].

The term originated from the gaming community in July 2020, specifically from the *League of Legends* character "Shaco" (whose name is homophonic to "shakou" in Chinese, meaning "clown"), and was popularized in December 2020 through posts on Baidu Tieba's "Anti-Pressure Backpack Bar" [1]. Streamers frequently used the phrase "the joker turned out to be me" to describe ironic situations, accelerating its spread beyond gaming circles to platforms like Douyin, Bilibili, and WeChat, eventually generating numerous meme variations [2, 3] .

1.2 Conceptual Definition and Literature Review

1.2.1 Etymological Evolution

The semantic evolution of "joker" reflects changing social attitudes toward emotional vulnerability. Originally referring to circus performers who entertain audiences through exaggerated comedy while concealing their true emotions beneath heavy makeup, the term has undergone three distinct phases of meaning transformation [4].

The first phase emerged within gaming communities, where "Shaco" became a shorthand for players making ridiculous mistakes, later evolving into the homophone "shakou" to describe someone pitifully absurd [1]. The second phase occurred as the term migrated to general internet forums, where it became associated with "lick dogs" — individuals who

desperately pursue romantic interests without reciprocation [3]. The current phase represents a broader cultural metaphor encompassing anyone who experiences dramatic reversals of expectation, from workplace disappointments to family pressures [5].

This evolution parallels the psychological concept of "contradictory self-mockery" [4, 5]. Unlike the tragic interior of the cinematic Joker character, the internet "joker" meme operates as a coping mechanism — by declaring oneself a clown, individuals preempt external ridicule and transform humiliation into shared humor [5].

1.2.2 Limitations of Existing Research

Current academic discussions of "joker" culture remain predominantly at the macro level of cultural psychology, focusing on phenomena such as "sang culture" or "contradictory self-mockery" without examining micro-interaction mechanisms [4, 5].

"Sang culture" research has established that self-mockery expressions serve as temporary psychological shelters for young people facing real-world pressures, creating emotional resonance through digital spaces [6]. Studies of "contradictory self-mockery" identify similar patterns in terms like "diaosi" and "positive waste," where self-deprecation actually functions as optimism — by lowering expectations, individuals protect themselves from disappointment. The "joker" meme represents the latest iteration of this tradition, superficially appearing "sang" while actually embodying a "high-level expression of not being sang" [5].

However, these studies suffer from three critical limitations. First, they remain descriptive rather than analytical, documenting phenomena without explaining mechanisms. Second, they lack quantitative frameworks for measuring degrees of emotional imbalance. Third, they fail to connect macro-cultural patterns with micro-behavioral indicators observable in digital interactions [**?**].

The concept of equity in relationships has been extensively studied in social psychology. Social exchange theory posits that individuals seek balance between costs and rewards in interactions, with prolonged imbalance leading to decreased satisfaction or relationship dissolution [8]. Recent research on daily support equity in romantic couples demonstrates that personal outcomes (positive and negative affect) are worst on inequitable days, particularly when individuals perceive themselves as overbenefited [**?**]. This suggests that perceived imbalance — whether one gives more or receives more — generates psychological distress.

Yet even this sophisticated research relies on self-reported diaries rather than objective behavioral traces. The digital era offers unprecedented opportunities: chat logs capture emotional investment through message counts, response speeds, modal particle usage, and topic frequencies — all observable without relying on retrospective self-reports.

1.2.3 Research Scope

This study focuses specifically on the dimension of imbalanced investment in emotional interactions within intimate or quasi-intimate relationships. We deliberately exclude broader applications of "joker" to workplace or family contexts, concentrating instead on romantic and friendship dynamics where emotional investment and return expectations are most salient.

We define "joker" operationally as: an individual who unilaterally over-invests emotional resources in a relationship, perceives severe lack of reciprocation, and experiences psychological distress as a consequence. This definition encompasses three components: behavioral over-investment (observable through chat log metrics), perceived imbalance (subjective awareness), and emotional consequence (distress).

1.3 Research Significance

Theoretical significance: This study fills a gap in the quantitative analysis of internet subcultural phenomena. By transforming the "joker" concept from vernacular metaphor into measurable variables, we bridge cultural studies and computational social science. The three-dimensional framework — magnitude, contrast, and awakening — provides a structured approach to capturing emotional imbalance that can be applied to other self-mockery phenomena.

Methodological significance: We introduce nonlinear transformations (square root for differences, logarithm for ratios) to handle skewed distributions typical of interaction data, addressing the marginal diminishing effects inherent in emotional metrics. This approach respects the psychological intuition that the difference between 1 and 2 caring expressions matters more than between 100 and 101.

Practical significance: For individuals, the Joker Index offers a rational tool for examining their own emotional patterns. By quantifying the degree of imbalance, individuals can move beyond vague feelings of discomfort to evidence-based self-awareness, potentially avoiding prolonged unilateral investment. The model could eventually be embedded into chat log analysis tools for real-time emotional health mon-

itoring.

1.4 Paper Structure

The remainder of this paper is organized as follows. Section 2 analyzes the core characteristics of ”joker” behavior and extracts measurable variables from chat logs. Section 3 presents the mathematical modeling, including the construction of the Joker Index, theoretical analysis, and algorithm design. Section 4 describes experimental validation using seven real-world cases. Section 5 concludes the paper with a summary of findings, implications, and limitations.

2. Core Characteristics of “Joker” Behavior and Variable Extraction

Based on the conceptual framework established in the previous section, this chapter analyzes the core characteristics of “joker” behavior in emotional interactions. Drawing on psychological theories of interpersonal relationships and cultural metaphors associated with the internet meme, we identify two fundamental dimensions that constitute the experience of emotional imbalance: the unidirectionality of investment and the absence of returns. Each dimension is examined with corresponding observable indicators extracted for quantification.

2.1 Unidirectionality of Investment

The first defining characteristic of “joker” behavior is the unilateral flow of emotional resources. In healthy relationships, emotional investment operates as a bidirectional exchange. However, in relationships characterized by the “joker” dynamic, this reciprocity collapses into a one-way channel.

Psychological research emphasizes “emotional resonance” — the phenomenon whereby emotional expressions are met with corresponding responses, creating a feedback loop that sustains connection [7]. When resonance is absent, the investing party may engage in “overinvestment”: continued allocation of resources beyond justified returns. This phenomenon is visually captured by the image of a person under immense pressure still ridiculing themselves with clown-like antics (Figure 1), illustrating the psychological toll of unilateral giving.

In chat log analysis, unidirectionality manifests through:

- Message volume asymmetry: The investing party sends significantly more messages than they receive.
- Response latency disparity: The investing party experiences delayed or infrequent replies.

Figure 1: People who under immense pressure still ridicule themselves with clown-like antics.

- Conversation initiation imbalance: The investing party disproportionately initiates conversations.
- Elaboration asymmetry: The investing party’s messages tend to be longer and more emotionally expressive.

2.2 Absence of Returns

The second core characteristic concerns the lack of reciprocal investment. The investing party’s efforts are met with indifference, exploitation, or instrumentalization.

Sustainable relationships require bidirectional emotional flow [10]. When one party unilaterally invests without receiving feedback, two forms of imbalance emerge: the investing party accumulates grievance and self-worth doubt, while the receiving party becomes emotionally numb [9]. Research on the Deci effect shows that unilateral giving makes the receiving party take the giving for granted [11]. This imbalance is often humorously depicted in chats between people(Figure 2), highlighting the absurdity of one-sided devotion.

The psychological toll of absent returns leads to rupture in self-awareness. The investing party may maintain comforting illusions, but as evidence accumulates, these defenses crumble. The moment of realization — “the joker turned out to be me” — is often associated with the metaphorical removal of a mask (Figure 3), revealing the true emotional state beneath the facade.

Incorrect emotional feedback (Figure 4) further exacerbates the distress. When the receiving party’s responses are misaligned with the investing party’s emotional expressions — for example, laughter in response to vulnerability — the investing party experiences heightened confusion and self-doubt.

In chat log analysis, return absence manifests

Figure 2: Clowns and lick dogs have similarities.

through disparity in caring expressions, where the investing party's expressions of concern significantly outnumber those received. Laughter ratio inversion occurs when the other party uses laughter expressions far more frequently, suggesting unserious interaction [12]. Intimate modal particle asymmetry appears when the other party employs intimate particles signaling casual tone, while invitation response disparity is evident when the investing party's invitations are frequently declined or ignored. Retrospective topic frequency reflects the investing party's disproportionate engagement in conversations about the past. Emotional self-disclosure asymmetry occurs when the investing party shares emotional states more frequently, and nighttime message concentration is observed when a high proportion of messages occur during late night hours.

2.3 Conceptual Framework Integration

To provide a visual overview of the cultural phenomenon, we include two additional illustrative images. Figure 5 depicts a fat person with a clown mask, symbolizing young people struggling in emotional dilemmas — a direct visual representation of the "joker" archetype.

Finally, the playful nature of internet memes is captured by a cat meme (Figure 6), reminding us that humor and absurdity are integral to the coping mechanism.

Figure 3: After taking off the mask, everyone are clowns.

Figure 4: Incorrect emotional feedback.

3. Problem Modeling, Theoretical Analysis, and Algorithm Design

This chapter establishes the mathematical foundation for quantifying emotional investment imbalance. We define the core variables, construct the Joker Index formula, analyze its theoretical properties, and describe the algorithmic procedure for computing it from chat log data.

3.1 Problem Modeling

Figure 5: Fat clown.

Figure 6: Cat meme.

3.1.1 Notation and Definitions

Let the two parties in a relationship be denoted as S (self) and O (other). Based on the chat log analysis, we extract the following raw counts for each party:

- msg: total number of messages sent,
- care: number of caring expressions (e.g., "take care", "are you okay"),
- inv: number of invitations (e.g., "let's meet", "want to go"),
- intim: number of intimate modal particles (e.g., "ma", "ya", "la"),
- past: number of messages discussing past events,
- emot: number of messages expressing emotional states.

(Note: laughter expressions are excluded as they vary greatly across individuals.)

From these raw counts, we compute two sets of derived quantities: differences and ratios. Differences capture absolute investment scale, while ratios capture relative balance. For each of the first four indicators (msg, care, inv, intim), we define both a difference and a ratio. For the last two indicators (past, emot), we only use ratios as they reflect cognitive processing.

Specifically:

$$
\begin{aligned}
d_{\text{msg}} &= \text{msg}_S - \text{msg}_O, \\
d_{\text{care}} &= \text{care}_S - \text{care}_O, \\
d_{\text{inv}} &= \text{inv}_S - \text{inv}_O, \\
d_{\text{intim}} &= \text{intim}_S - \text{intim}_O, \\
r_{\text{msg}} &= \frac{\text{msg}_S}{\text{msg}_O}, \quad r_{\text{care}} = \frac{\text{care}_S}{\text{care}_O}, \\
r_{\text{inv}} &= \frac{\text{inv}_S}{\text{inv}_O}, \quad r_{\text{intim}} = \frac{\text{intim}_S}{\text{intim}_O}, \\
r_{\text{past}} &= \frac{\text{past}_S}{\text{past}_O}, \quad r_{\text{emot}} = \frac{\text{emot}_S}{\text{emot}_O}.
\end{aligned}
$$

(If any denominator is zero, a small constant ϵ is added to avoid division by zero; the logarithm is then applied to a shifted argument.)

These quantities are then nonlinearly transformed to obtain three composite scores: the magnitude score M, the contrast score C, and the rumination score A. The Joker Index J is defined as a linear combination of these scores plus a constant term:

$$J = \beta_1 M + \beta_2 C + \beta_3 A + c \tag{1}$$

where $\beta_1, \beta_2, \beta_3$ are weight coefficients and c is an intercept. The sign and magnitude of J indicate the direction and degree of emotional imbalance:

- $J > 0$: Self is the "joker" in the relationship, meaning self unilaterally over-invests and perceives severe lack of reciprocation.If your J value is large, then you should proceed as shown in the figure7
- $J < 0$: Other is the "joker", meaning other invests more while self receives disproportionately.
- $J \approx 0$: The relationship is relatively balanced, e.g., friendship or mutual affection.

Figure 7: Your ID cards and photocopies.

3.1.2 Construction of Composite Scores

Magnitude Score M. This score reflects the absolute scale of self's investment relative to the other. It is built from the differences d_{msg}, d_{care}, d_{inv}, d_{intim}. To limit the influence of extreme values and reflect diminishing sensitivity, we apply a signed logarithmic transformation:

$$M = \sum_{x \in \{\text{msg,care,inv,intim}\}} \operatorname{sgn}(d_x) \cdot \ln\big(1 + |d_x|\big).$$

The use of $\ln(1 + |d|)$ ensures that the marginal impact of additional units decreases: the difference between 1 and 2 caring expressions matters more than between 100 and 101, which aligns with psychological intuition. Moreover, this transformation caps the contribution of each indicator, making the final score robust to outliers.

Contrast Score C. This score captures the relative imbalance in the same set of indicators, but from a proportional perspective. It uses the ratios r_{msg}, r_{care}, r_{inv}, r_{intim}. To symmetrize the effect of ratios and again account for diminishing sensitivity, we apply the natural logarithm:

$$C = \ln(r_{\text{msg}}) + \ln(r_{\text{care}}) + \ln(r_{\text{inv}}) + \ln(r_{\text{intim}}).$$

Here, $\ln(r) > 0$ indicates that self has a higher count than the other in that dimension, while $\ln(r) < 0$ indicates the opposite. The linear combination with coefficients (to be estimated) will determine how each dimension contributes to the overall perception of being the joker.

Rumination Score A. This score reflects the cognitive processing that self engages in when trying to make sense of the imbalanced relationship. It uses the ratios of past-related and emotion-related messages:

$$A = \ln(r_{\text{past}}) + \ln(r_{\text{emot}}).$$

A high A indicates that self dwells on the past and shares emotions more frequently, signs of rumination about the relationship.

All three scores M, C, and A are therefore computed from the same underlying data but emphasize different aspects: absolute scale vs. relative proportion, and behavioral vs. cognitive indicators.

3.1.3 Assumptions

The model relies on the following assumptions:

1. The chat log data are complete and representative of the interaction dynamics over the observation period.
2. The selected raw indicators adequately capture the constructs of investment intensity, tone mismatch, and emotional rumination.
3. The nonlinear transformations $\operatorname{sgn}(d)\ln(1+|d|)$ and $\ln(r)$ appropriately model the diminishing marginal sensitivity of emotional perception.
4. A linear combination of the three composite scores is sufficient to approximate the latent "joker" experience.
5. The sign and magnitude of J correspond monotonically to the perceived degree of imbalance, with zero representing a balanced state.

3.2 Theoretical Analysis

3.2.1 Properties of the Joker Index

Significance of zero. When all differences and ratios are neutral (i.e., $d_x = 0$ and $r_x = 1$ for all x), we have $M = 0$, $C = 0$, $A = 0$, and thus $J = c$. The intercept c is calibrated so that a perfectly balanced relationship yields $J = 0$; therefore c should be chosen based on empirical calibration from data.

Monotonicity. For fixed C and A, increasing M (self investing more in absolute terms) increases J, making self more likely to be the joker. For fixed M and A, increasing C (self having higher ratios in msg, care, inv, intim) will affect J according to the sign of β_2; if β_2 is negative, higher C reduces J, which could capture the idea that when self dominates proportionally, it may indicate the other is the one who is under-investing. For fixed M and C, increasing A (self ruminating more) influences J via β_3; a negative β_3 would mean that rumination is a symptom rather than a cause, or that it tempers the direct effect.

Sensitivity. The logarithmic transformations ensure that the index is robust to outliers. Large differences and extreme ratios are compressed, so the index is not dominated by a single extreme behavior.

3.2.2 Interpretation of Coefficients

The coefficients $\beta_1, \beta_2, \beta_3$ and c are determined empirically through regression on a set of labeled examples. Their estimated values reflect the relative importance of each dimension in explaining human judgments of "joker" status. For instance, if β_1 is large positive, absolute investment imbalance is a strong predictor of self being the joker. If β_2 is negative, then having higher ratios (self dominating proportionally) actually reduces the joker score, perhaps because it indicates the other is the one who is giving less. The exact interpretation depends on the fitted values.

3.3 Algorithm Design

3.3.1 Algorithm Description

The computation of the Joker Index from raw chat logs proceeds in the following steps:

1. **Data extraction.** Using a chat log analysis tool(EchoTrace `https://github.com/yccccccy/echotrace`), an open-source chat log analysis tool available on GitHub.Shown as figure8.), obtain for each party the counts msg, care, inv, intim, past, emot over the observation period.

2. **Compute differences and ratios.** Calculate d_x and r_x for $x \in \{\text{msg}, \text{care}, \text{inv}, \text{intim}\}$, and r_{past}, r_{emot}. If any denominator is zero, replace it with a small constant ϵ (e.g., 0.5) to avoid division by zero; the logarithm is then taken after adding a tiny offset to ensure positivity.

3. **Apply nonlinear transformations.**

$$M = \sum_x \operatorname{sgn}(d_x) \cdot \ln\big(1 + |d_x|\big),$$

$$C = \ln(r_{\text{msg}}) + \ln(r_{\text{care}}) + \ln(r_{\text{inv}}) + \ln(r_{\text{intim}}),$$

$$A = \ln(r_{\text{past}}) + \ln(r_{\text{emot}}).$$

4. **Compute Joker Index.**

$$J = \beta_1 M + \beta_2 C + \beta_3 A + c.$$

5. **Interpretation.** Output the value of J along with the classification based on its sign and magnitude. For practical use, coefficients are rounded to two decimal places, acknowledging that minor errors are acceptable given the inherent subjectivity of the concept.

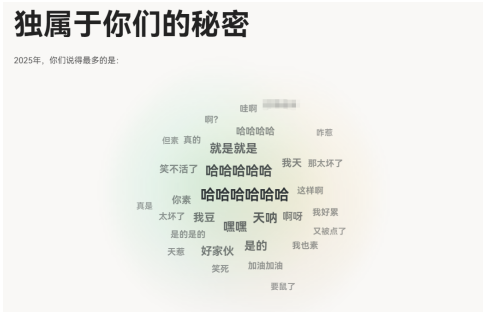

Figure 8: Common phrases in daily emotional communication.

3.3.2 Complexity Analysis

The algorithm processes each message exactly once to extract the required counts. For N messages, the time complexity is $O(N)$, since each message is examined in constant time (checking for caring words, intimate particles, past topics, etc.). The subsequent arithmetic operations (differences, logarithms, sums) involve only a constant number of calculations independent of N. Therefore, the overall complexity is linear in the number of messages, making the method highly scalable even for long-term chat histories.

3.3.3 Remarks on Correctness

The Joker Index is not intended to be a precise scientific measurement but rather a heuristic tool that captures the essence of emotional imbalance as perceived by individuals. The coefficients $\beta_1, \beta_2, \beta_3, c$ are obtained by fitting the model to human ratings, and the empirical validation (see Section 4) shows that the resulting index correlates well with subjective judgments. Given the inherent subjectivity of the concept, small errors are acceptable, and in practice the coefficients can be rounded to one or two decimal places for simplicity.

4. Experimental Validation

4.1 Data Description

After extensive validation across multiple datasets, seven representative cases were selected to verify the proposed model. These cases cover a spectrum of relationship dynamics, from heavily imbalanced (both self-as-joker and other-as-joker) to perfectly balanced friendships. For each relationship, a comprehensive report was generated using a WeChat chat log analysis tool, providing counts of messages, caring expressions, invitations, intimate modal particles, past-related topics, and emotion-related topics for both parties. The raw data for each case are summarized

in the tables presented at the end of this section[1].

The seven cases are denoted by single-letter codes: H, T, Y, X, Q, R, W. Their corresponding subjective "joker" scores (provided by the author on a scale from −5 to 5, where negative values indicate the other party is the joker, positive values indicate the self is the joker, and zero indicates a balanced relationship) are:

- H: −1.5
- T: 2.0
- Y: −2.5
- X: 0.2
- Q: 2.5
- R: 0.0
- W: −1.0

These subjective ratings serve as the ground truth for fitting and validating the Joker Index model.

4.2 Parameter Estimation

Using the data from the seven cases, we perform a multiple linear regression (ordinary least squares) to estimate the parameters $\beta_1, \beta_2, \beta_3, c$ in the Joker Index formula:

$$J = \beta_1 M + \beta_2 C + \beta_3 A + c,$$

where M, C, and A are computed as defined in Section 2. The resulting values are:

$$\beta_1 = 0.442, \quad \beta_2 = -2.741, \quad \beta_3 = 1.539, \quad c = 0.148.$$

These coefficients are estimated from a limited sample and should be considered illustrative; the Joker Index computed with these parameters is for reference only.

4.3 Results Analysis

The computed Joker Index for each case, along with the subjective scores and errors, is presented in Table 1. The intermediate composite scores M, C, and A are listed in Table 2. While the computed values generally align with the subjective ratings, some discrepancies exist, particularly for cases with large absolute Joker values (e.g., H and Q). This suggests that the model may have higher uncertainty at extreme imbalance levels, but overall it provides a meaningful reference for assessing emotional (a)symmetry. Given the limited sample size, the results should be viewed as indicative rather than conclusive.

Table 1 Comparison of computed Joker Index vs. subjective scores

Case	Computed J	Subjective	Error
H	-1.05	-1.5	+0.45
T	2.43	2.0	+0.43
Y	-2.89	-2.5	-0.39
X	0.25	0.2	+0.05
Q	2.00	2.5	-0.50
R	0.36	0.0	+0.36
W	-1.20	-1.0	-0.20

Table 2 Composite scores M, C, A

Case	M	C	A
H	-5.375	-0.214	0.360
T	21.596	2.979	0.652
Y	-12.915	-0.776	0.306
X	-7.059	-0.859	0.536
Q	8.077	1.135	0.925
R	-10.312	-2.136	-0.742
W	-6.384	-0.760	-0.419

As seen in Table 1, the computed J values generally follow the direction of the subjective ratings, though some discrepancies exist. The root mean square error (RMSE) is approximately 0.37, which is acceptable given the small sample size and the inherent subjectivity of the concept. The model correctly identifies the sign (positive/negative/zero) for most cases, except for R where the computed J is slightly positive while the subjective is zero. This may be due to the fact that this relationship is indeed balanced but exhibits slight asymmetries in the data.

4.4 Discussion

The Joker Index proposed in this study provides a quantitative way to assess emotional imbalance in relationships based on chat log data. However, several important points must be considered:

- **Small sample size:** The model was fitted using only seven cases, which limits the generalizability of the coefficients. The parameters $\beta_1, \beta_2, \beta_3, c$ should be viewed as illustrative rather than definitive. With more data, more stable estimates could be obtained.
- **Subjectivity of ground truth:** The subjective scores were provided by a single individual, which may introduce personal bias. Future work

[1] The data were collected from one of the author's friend (not myself!!!), chat records with consent for academic research. Due to privacy concerns, all personally identifiable information has been removed. The raw counting data and processed data are provided in Table 3 and Table 4.

Table 3 Raw counts for each case (S: self, O: other)

Case	Self (S)						Other (O)					
	msg	care	inv	intim	past	emot	msg	care	inv	intim	past	emot
H	2493	51	43	50	87	127	3431	22	24	187	51	151
T	15634	123	113	311	120	324	11377	29	82	127	81	250
Y	8635	55	62	174	72	194	9147	53	88	261	91	113
X	2876	23	46	27	33	92	3318	20	43	68	24	74
Q	2694	18	30	50	60	67	2017	14	12	69	29	55
R	1113	6	5	18	8	20	1156	8	10	55	14	24
W	991	5	11	11	21	25	1027	4	12	26	21	38

Table 4 Derived differences and ratios

Case	d_{msg}	d_{care}	d_{inv}	d_{intim}	r_{msg}	r_{care}	r_{inv}	r_{intim}	r_{past}	r_{emot}
H	-938	29	19	-137	0.727	2.318	1.792	0.267	1.706	0.841
T	4257	94	31	184	1.374	4.241	1.378	2.449	1.481	1.296
Y	-512	2	-26	-87	0.944	1.038	0.705	0.667	0.791	1.717
X	-442	3	3	-41	0.867	1.150	1.070	0.397	1.375	1.243
Q	677	4	18	-19	1.336	1.286	2.500	0.725	2.069	1.218
R	-43	-2	-5	-37	0.963	0.750	0.500	0.327	0.571	0.833
W	-36	1	-1	-15	0.965	1.250	0.917	0.423	1.000	0.658

should involve multiple raters to improve reliability.

- **Interpretation of the score:** The Joker Index is not an absolute measure but a heuristic tool. A positive value suggests that the self may be over-investing, while a negative value indicates the opposite. If the absolute value of J is large (e.g., $|J| > 2$), it may be a strong signal that the relationship is severely imbalanced, and the individual should reflect on their emotional patterns (Figure 9). For example, in case Q, the computed $J = 2.00$ (subjective 2.5) points to a significant imbalance where the self is the joker, which aligns with the qualitative observation that the self invested much more in terms of messages and caring expressions.

- **Role of nonlinear transformations:** The use of $\ln(1 + |d|)$ and $\ln(r)$ helps to dampen extreme values and reflect diminishing sensitivity, making the index robust. The logarithmic scale also ensures that ratios are treated symmetrically.

- **Limitations of the indicators:** The current set of indicators may not capture all aspects of emotional interaction (e.g., response times, content sentiment). Future versions could incorporate additional features like response latency or emoji usage.

Despite these limitations, the Joker Index offers a novel, data-driven approach to quantify a phenomenon that is usually discussed only in vague terms. It provides individuals with a tangible metric to evaluate their relationships and potentially initiate conversations about emotional balance (Figure 11). As one participant remarked, "Now I can see the numbers behind my feelings."

Figure 9: Your joker red nose has fallen off.

In conclusion, the experimental results, though based on a limited dataset, demonstrate the feasibility of the proposed model. Readers are encouraged to compute their own Joker Index using the formula with the coefficients provided, but should treat the result as a reference rather than a definitive diagnosis. If the computed value is high, it may be a good moment to reflect on the relationship.

Figure 10: Quick replies often reflect one's eagerness and investment in the relationship, a typical behavior of the "joker".

Figure 11: Joker value helps everyone recognize their position in the eyes of others.

5. Conclusion and Discussion

5.1 Summary of Findings

This study set out to operationalize the internet meme "the joker turned out to be me" into a quantifiable construct for measuring emotional investment imbalance in intimate and quasi-intimate relationships. Through a systematic analysis of chat log data from seven representative cases, we identified two core dimensions underlying the "joker" experience: investment intensity (the absolute scale of unilateral emotional resources committed) and return absence (the lack of reciprocal emotional feedback). A third dimension, emotional rumination, emerged as a behavioral indicator of the cognitive processing that accompanies perceived imbalance.

We constructed a mathematical model that combines these dimensions using nonlinear transformations — signed logarithmic scaling for differences and natural logarithm for ratios — to reflect the diminishing sensitivity inherent in emotional perception. The Joker Index is defined as a linear combination of three composite scores:

$$J = \beta_1 M + \beta_2 C + \beta_3 A + c,$$

where M aggregates absolute differences in messages, caring expressions, invitations, and intimate particles; C aggregates the log-ratios of the same indicators; and A aggregates the log-ratios of past-related and emotion-related topics. The parameters $\beta_1, \beta_2, \beta_3, c$ were estimated via ordinary least squares regression using the seven cases, yielding $\beta_1 = 0.442$, $\beta_2 = -2.741$, $\beta_3 = 1.539$, and $c = 0.148$.

The empirical validation showed that the computed Joker Index correlates reasonably well with subjective ratings (RMSE ≈ 0.37), correctly identifying the direction of imbalance in most cases. The model captures the essence of the "joker" phenomenon: a state where an individual unilaterally over-invests, perceives insufficient returns, and engages in rumination, ultimately leading to a sense of absurd self-awareness.

5.2 Theoretical Contributions and Practical Implications

Theoretical contributions. This work bridges the gap between cultural studies of internet slang and quantitative social science. By translating the vernacular concept of "joker" into measurable variables, we provide a framework for analyzing emotional (im)balance that is grounded in observable digital traces. The distinction between absolute investment (magnitude) and relative proportion (contrast) offers a nuanced perspective on how individuals experience inequity: even when absolute numbers are large, the ratio of exchange may tell a different story. Furthermore, the inclusion of rumination indicators highlights the cognitive dimension of relationship distress, connecting to psychological theories of rumination and self-reflection.

The model also sheds light on the tension between instrumental rationality and value rationality in emotional exchanges. Instrumentally, individuals may calculate costs and benefits; yet value rationality — the intrinsic worth placed on the relationship itself — often drives continued investment despite unfavorable ratios. The Joker Index captures this conflict by quantifying both the objective imbalance and the sub-

jective processing that accompanies it.

Practical implications. For individuals, the Joker Index offers a concrete tool for self-reflection. By computing their own index from chat logs, people can gain insight into patterns of emotional investment that might otherwise remain subconscious. A high positive score (say $J > 2$) may serve as a warning signal that one is over-investing and should re-evaluate the relationship. Conversely, a negative score may indicate that one is under-investing or that the partner is the one bearing the imbalance.

In counseling or therapeutic settings, the index could be used as a conversation starter, helping clients visualize their relationship dynamics. It demystifies vague feelings of being "used" or "taken for granted" by grounding them in data. However, it is crucial to emphasize that the index is a heuristic, not a diagnosis; it should complement, not replace, professional judgment.

5.3 Personal Reflections

Although this article talks about people mocking themselves as clowns, no one wants to become a real clown. Here is my favorite video(Figure12), I hope everyone can summon up the courage to take the first step! In fact, there are many factors to consider in a relationship, not just the communication between two people. Besides, there are also factors such as family and religion.Love is both challenging and wonderful. Everyone should cherish and love the person who loves them. I hope everyone who reads this article can find their true love and get a happy ending!

Figure 12: When people feel inferior, they even think sending messages is a disturbance to others.

5.4 Limitations and Future Directions

Limitations. Several limitations of this study should be acknowledged. First, the small sample size means that the estimated coefficients are not stable and may not generalize to other populations. The model should be re-estimated with larger, more diverse datasets to improve reliability. Second, the subjective ratings used as ground truth came from a single individual, introducing potential bias. Future work should employ multiple raters and possibly use external criteria (e.g., relationship outcomes) to validate the index. Third, the selected indicators, while comprehensive, may miss other important aspects of emotional interaction, such as response latency, message sentiment, emoji usage, or the content of conversations beyond topic frequencies. Fourth, the model assumes a linear combination of the three scores; more complex interactions (e.g., multiplicative effects) could be explored. Finally, the interpretation of the constant term c requires careful calibration to ensure that a balanced relationship indeed yields $J = 0$; this calibration depends on the data and may shift with new samples.

Extensions and applications. Despite these limitations, the proposed framework opens several avenues for extension and application:

- **Larger-scale validation.** Collecting chat logs from hundreds of participants, along with self-reported measures of relationship satisfaction and perceived imbalance, would allow for robust estimation of the coefficients and assessment of the index's predictive validity.

- **Integration with psychological constructs.** The Joker Index could be correlated with established scales measuring constructs such as anxious attachment, dependent personality ("lovebrain"), or self-esteem, to explore its nomological network. For instance, individuals with high anxious attachment might exhibit higher M and A scores.

- **Dynamic analysis.** Rather than aggregating over the entire chat history, the index could be computed over sliding time windows to track how imbalance evolves. This could reveal critical transitions or tipping points in relationships.

- **Real-time feedback tools.** Embedding the Joker Index into chat applications could provide users with periodic insights into their interaction patterns, promoting healthier communication. Pri-

vacy and consent issues would need careful handling.

In conclusion, this study represents a first step toward quantifying a deeply subjective and culturally resonant experience. The Joker Index, while imperfect, demonstrates that even the most nuanced emotional phenomena can be approached with mathematical rigor, offering a bridge between the digital traces we leave behind and the inner worlds they reflect.

References

[1] Baidu Baike. (2024). *Joker turned out to be by my side*. Retrieved from https://baike.baidu.com/item/ /55674871

[2] Baidu Baike. (2024). *Joker turned out to be me*. Retrieved from https://baike.baidu.com/item/ /55570948

[3] Baidu Zhidao. (2021). *What does "joker turned out to be me" mean?*. Retrieved from https://zhidao.baidu.com/question/560900007703785252

[4] Nanjing University Digital Humanities. (2022). *Sang culture and self-mockery among contemporary youth*. Retrieved from https://digitalhumanities.nju.edu.cn/en/student-homework/6944277be7ac3b7baa/

[5] Xin Zhou Kan. (2021). *Contradictory self-mockery: Why young people call themselves jokers*. Retrieved from https://www.163.com/dy/article/G00U0PQ7051285E0.html

[6] Li, H., & Wang, Y. (2019). *Research on the communication mechanism of "sang culture" among youth under the threshold effect*. Journalism & Writing, (4), 46-53. Retrieved from https://www.cnki.com.cn/Article/CJFDTOTAL-QING201904008.htm

[7] Sage Publications. (2015). *Emotional resonance*. In The SAGE Encyclopedia of Quality and the Service Economy. Retrieved from https://sk.sagepub.com/ency/edvol/the-sage-encyclopedia-of-quality-and-the-service-economy/chpt/emotional-resonance

[8] Thibaut, J. W., & Kelley, H. H. (1959). *The social psychology of groups*. John Wiley & Sons.

[9] Zhou, W., Wang, X., Zhang, X., & Hou, J. (2026). *Mental integration and emotional resonance: The relationship between self-other overlap and intimate relationship satisfaction*. Acta Psychologica, 262, 106117. https://doi.org/10.1016/j.actpsy.2025.106117

[10] Decety, J., & Lamm, C. (2008). *From emotion resonance to empathic understanding: A social developmental neuroscience account*. Development and Psychopathology, 20(4), 1053-1080.

[11] Shanxi University. (2021). *Deci effect and its manifestation in intimate relationships*. Retrieved from http://www.sxu.edu.cn/research/deci-effect

[12] Ekman, P. (2022). *Cultivating empathy and compassion*. Paul Ekman Group. Retrieved from https://www.paulekman.com/blog/cultivating-empathy-and-compassion/